Clase 7 - Fundamentos de Ingeniería de Datos

Objetivos de la Clase

- Make an ***Introduction*** to data engineering
- Discuss the ***Characteristics of Data***
- What are ***Modern Data Platforms***?
- Take our first steps in data pipeline ***Orchestration***

Agenda

1. **Introduction**
2. Characteristics of Data
3. Modern Data Platforms
4. Orchestration

Introduction

“Data is the **new oil**” and other *stories*...

What are data products?

- An asset that **delivers value through data** (insights, APIs, reports, models, dashboards)
- Difference “project” vs. “product”: continuity, roadmap, SLAs.
- Attributes: **usefulness**, **reliability**, **maintainability**, **governance**.

Core motivation: Data-Driven Decisions

- What **products are relevant** to my users?
- What clients are more prone to **churn**?
- What is the **seasonality** of my business?
- Is this payment operation **fraudulent**?

Valuable insights are harder to produce



More valuable

Descriptive

What happened?

Diagnostic

Why did something happen?

Predictive

What will happen?

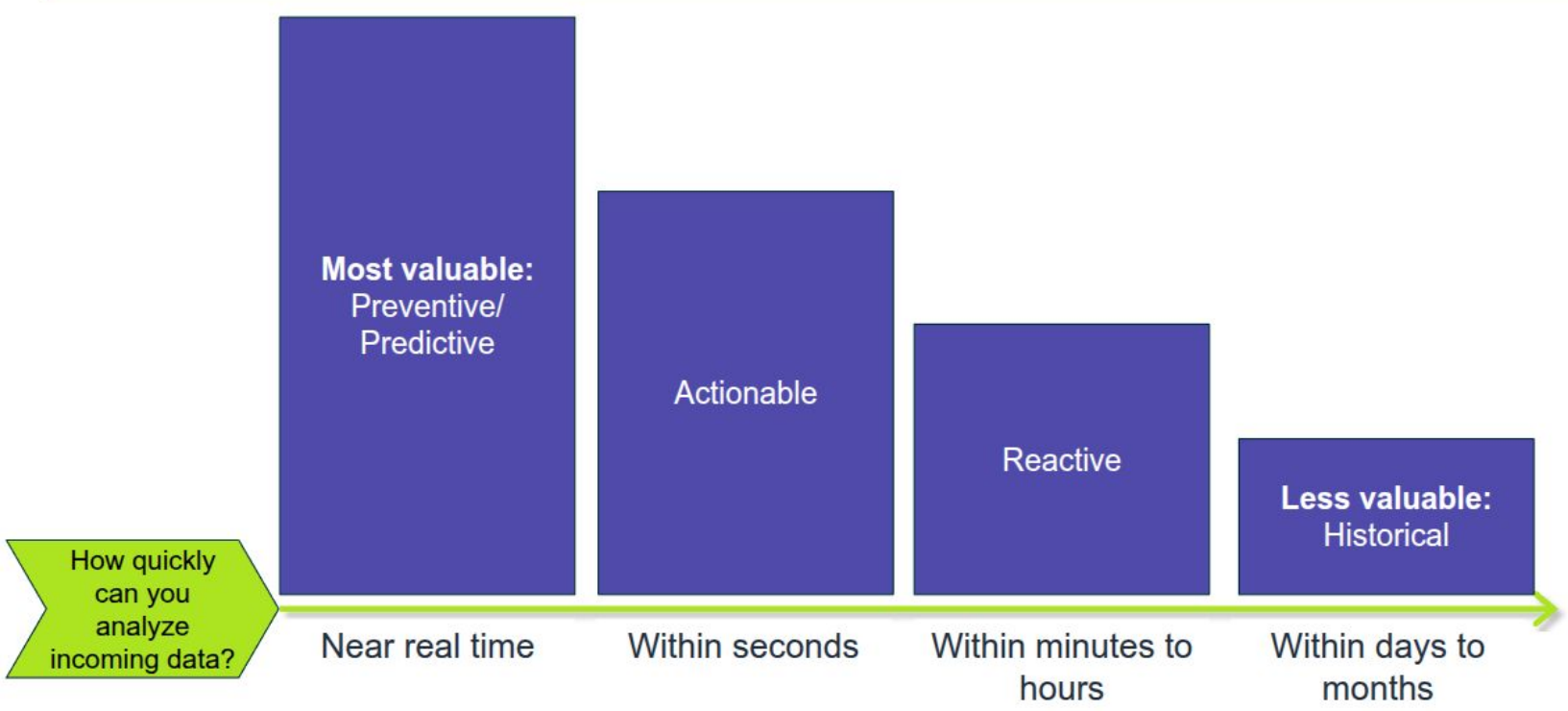
Prescriptive

How can we make something happen or prevent something from happening?

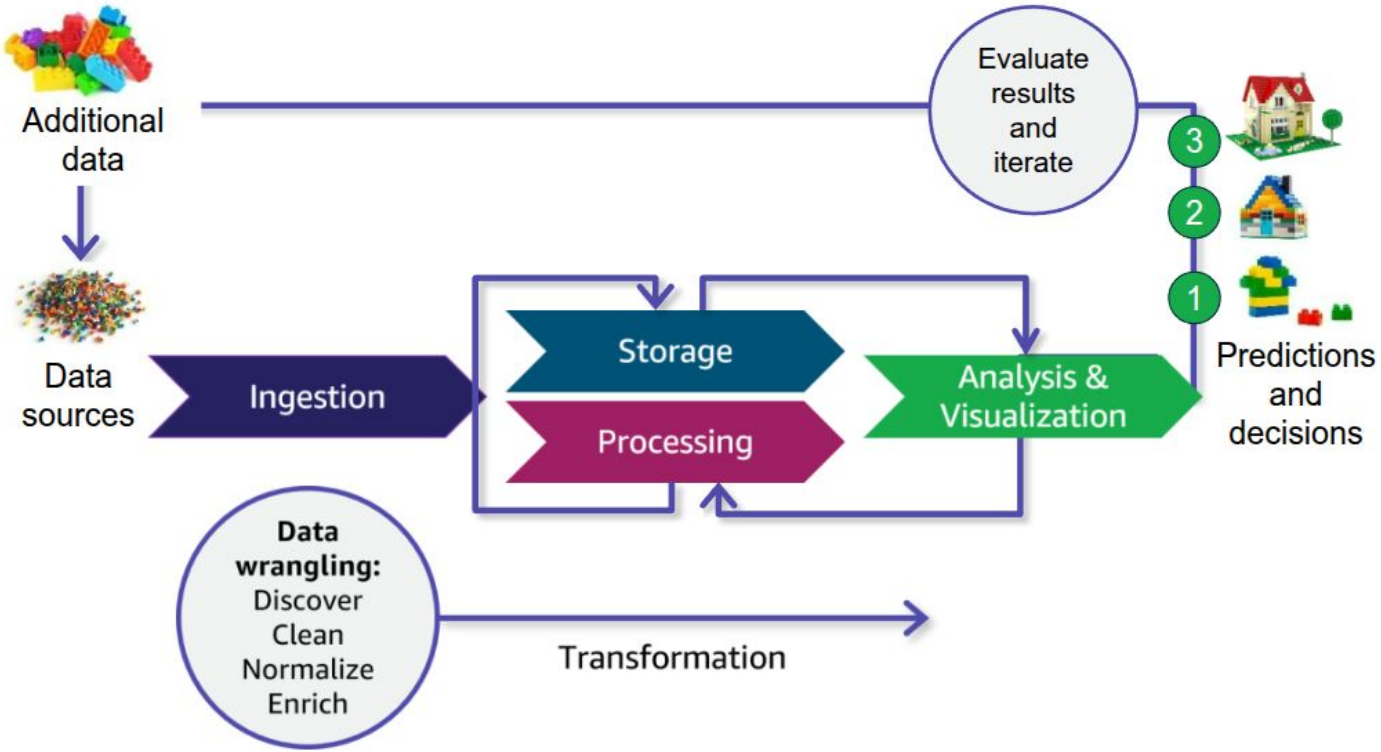
Less valuable, easier to derive

More difficult

Fresh insights are more valuable



Layers of the pipeline infrastructure



Trade-offs Everywhere



Cost

- How much should you invest to go faster or predict more accurately?
- How much incremental improvement justifies additional cost?



Speed

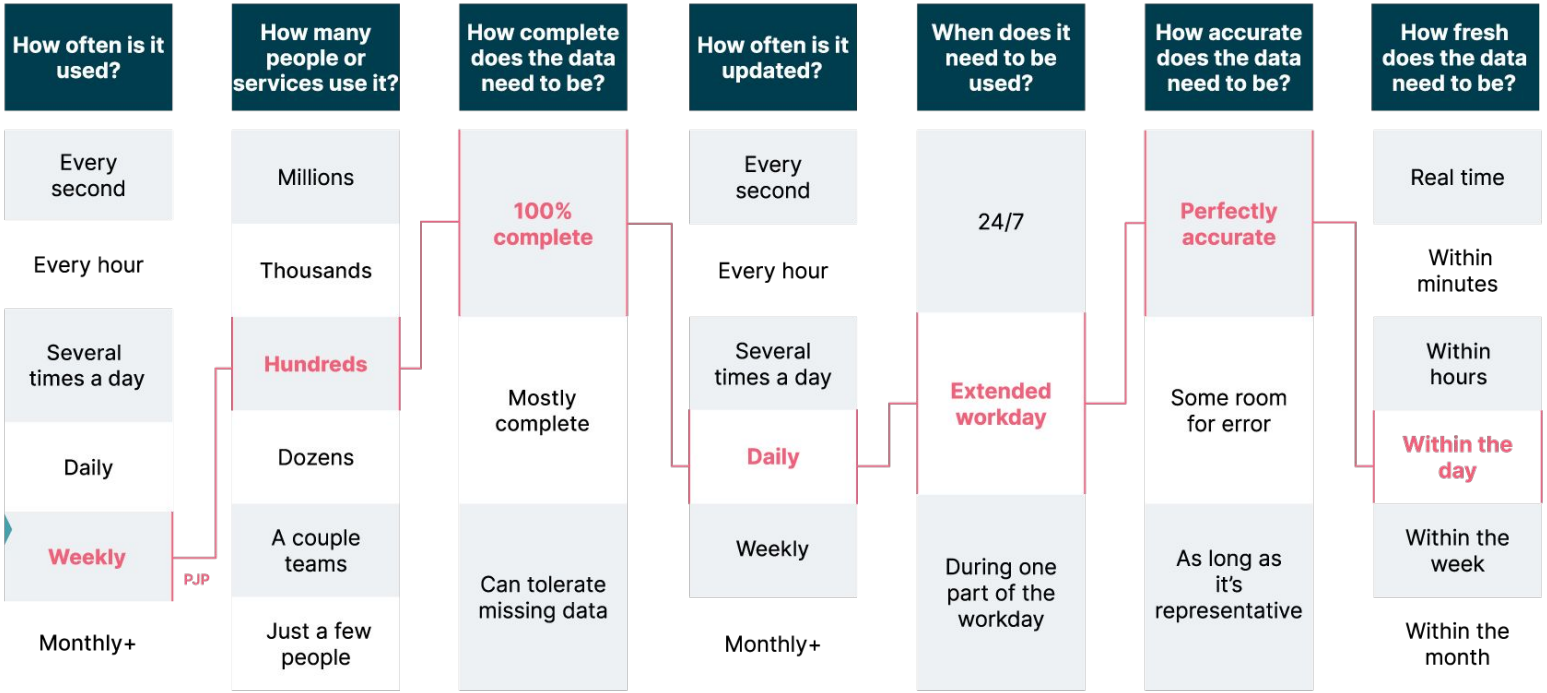
- How quickly do you need an answer?
- Can you sacrifice accuracy for speed?



Accuracy

- How accurate does the prediction need to be?
- Does waiting for a better answer outweigh answering more quickly?

Trade-offs Everywhere



Best practice: Working backwards



2. What data do you need to support this?

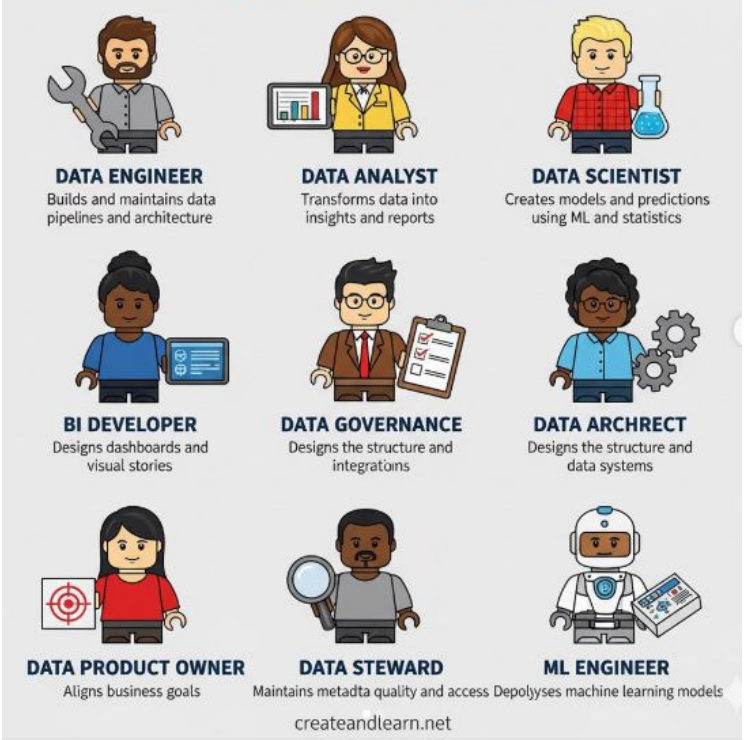
1. What decision are you trying to make?



Weigh the trade-offs of cost, speed, and accuracy.

Roles on a data team (overview)

- **Data Engineer**: enables movement/transformation/serving.
- **Analytics Engineer**: analytical modeling, semantics, lightweight tests.
- **Data Analyst**: exploration, BI, hypotheses and communication.
- **Data Scientist/ML Engineer**: experimentation, models, ML deployment.
- **Data Product Manager / Data PM**: discovery, prioritization, product KPIs.



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Data characteristics that drive infrastructure decisions



Volume

How big is the dataset? How much new data is generated?



Velocity

How frequently is new data generated and ingested?



Variety

What types and formats? How many different sources does the data come from?



Veracity

How accurate, precise, and trusted is the data?

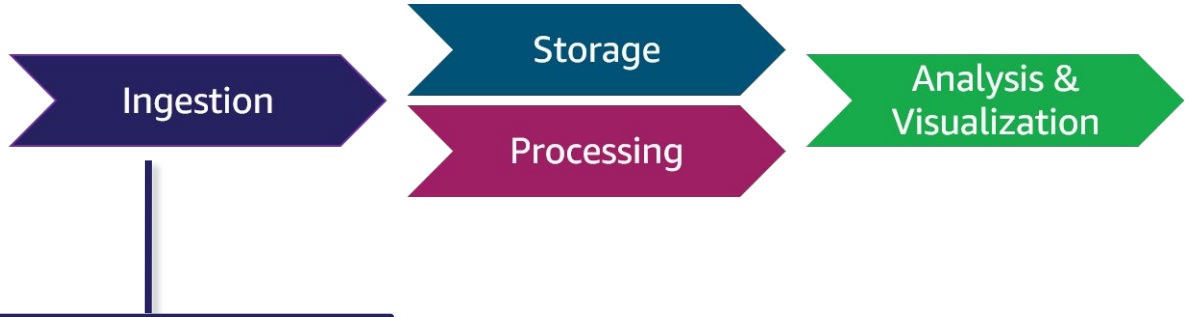


Value

What insights can be pulled from the data?

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Ingestion decisions related to volume and velocity



Examples:

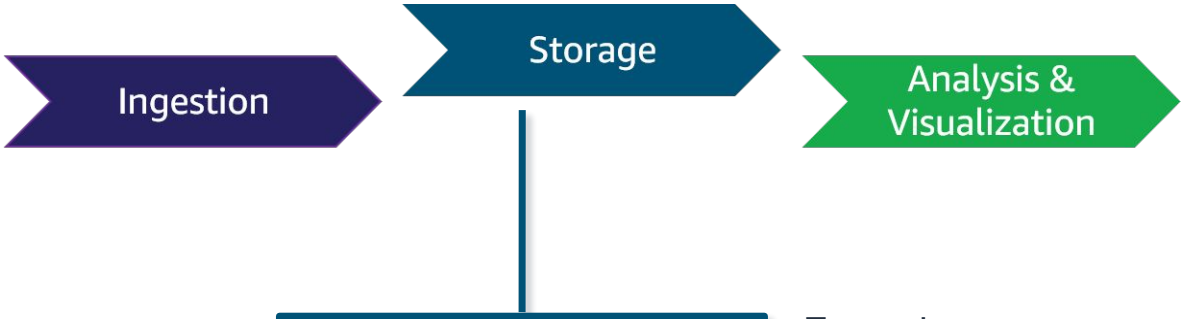
Streaming ingestion: Clickstream data from a retailer's website sends a large volume of small bits of data at a continuous pace. Analysis must be performed immediately.

Batch ingestion: Sales transaction data from retailers across the world is sent to a central location periodically. Data is analyzed overnight with reports sent to branches in the morning.

aws

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Storage decisions related to volume and velocity



Examples:

Long term, reporting access: Five years of sales data is stored for trending analysis, which is performed monthly.

Short term, very fast access: Incoming ecommerce sales transaction data is used to suggest additional purchases within the current session.

aws

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Processing decisions related to volume and velocity



How much data must be processed in a single iteration? Does it require a distributed solution? How quickly and how often does processing need to occur?

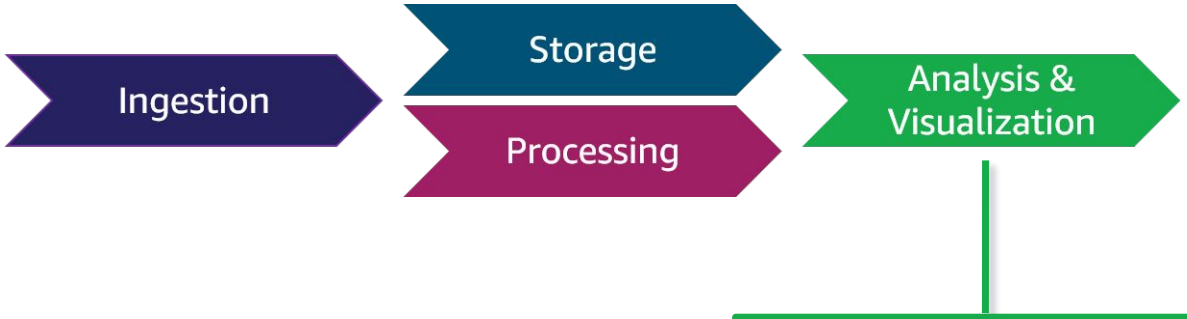
Examples:

Big data processing: Analytics is performed on credit card data for all US-based transactions in the past week.

Streaming analytics: Real-time alerts are produced on log data to identify potential fraud as soon as it occurs.

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Analysis/visualization decisions related to volume and velocity



Examples:

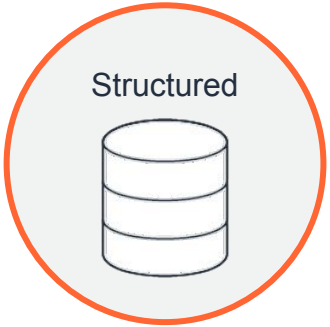
Historical analysis: A year's worth of sales data is visualized, and users can drill down by region and salesperson.

Streaming Internet of Things (IoT) data: The real-time error rates of sensors in a plant are visualized.

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Data types

Easier to use "Hotter"

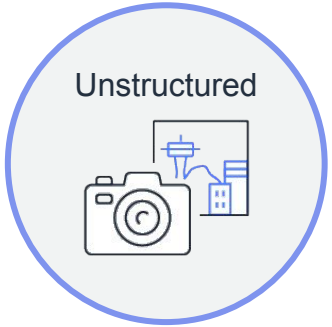


- Rows and columns
- Well-defined schema
- Example: Relational database tables



- Elements and attributes
- Self-describing structure
- Examples: CSV, JSON, XML

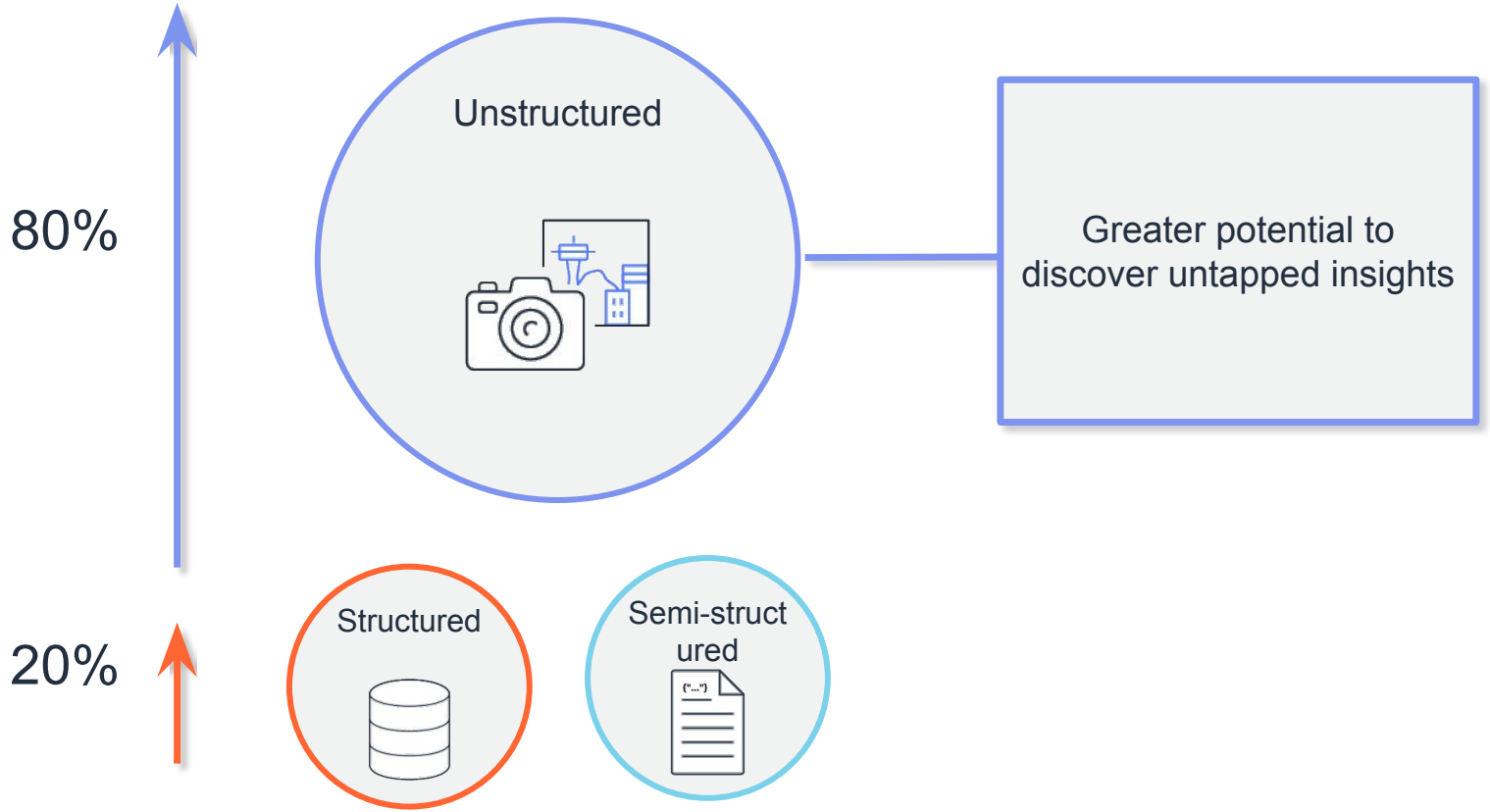
- Files
- No predefined structure
- Examples: Images, movies, clickstream data



More flexible "Colder"

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80% or more of available data is unstructured



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Example use cases for data types

Structured	Semistructured	Unstructured
Query a relational database to report on customer service tickets that were submitted in in a specific period.	Extract and analyze customer comments from an online chat application that saves conversations as JSON.	Perform sentiment analysis on customer service emails.

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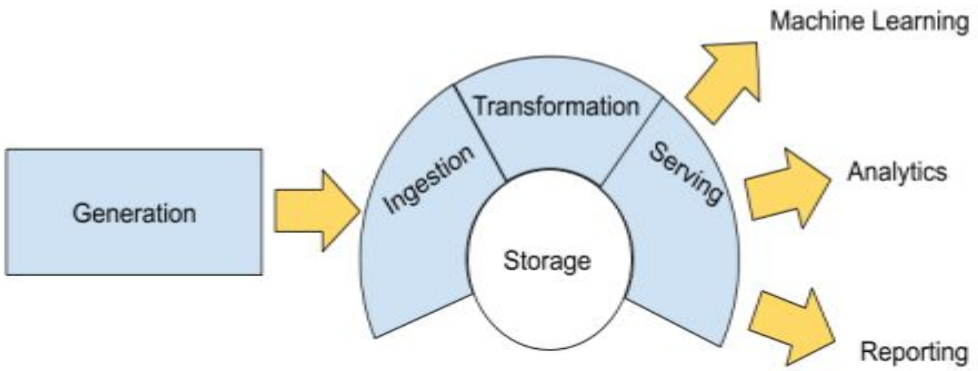
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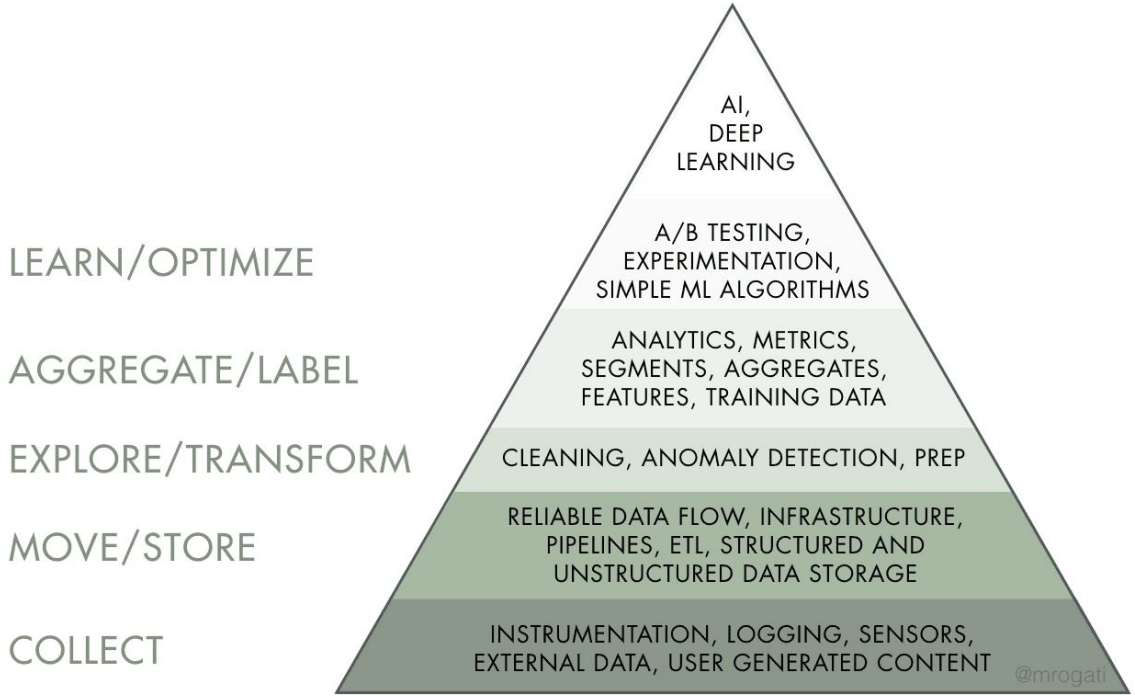
1. Introduction
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3. **Modern Data Platforms**
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Data Lifecycle

1. Generation/Capture
2. Ingestion
3. Storage
4. Transformation
5. Consumption

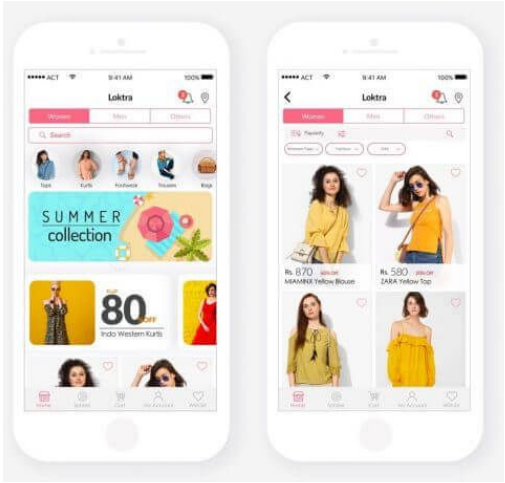


The data hierarchy of needs

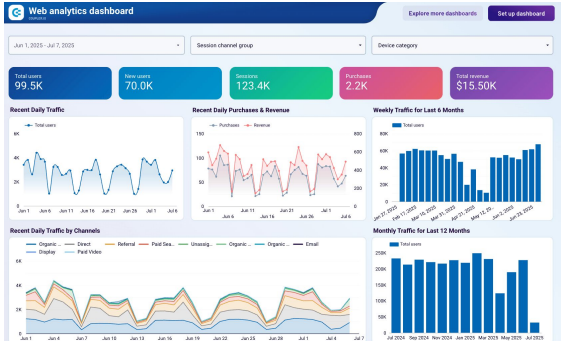


Databases in two flavors

- **OLTP** (Online **Transaction** Processing)
  - Handles real-time transactional data efficiently and reliably.



- **OLAP** (Online **Analytical** Processing)
  - Analyzes large data sets for insights and decision-making.



OLTP

- Manages and processes **real-time transactions**
  - Ej: Login, user settings, update user profile, items checkout, etc
- Uses **relational** databases.
- Uses **normalized** or denormalized models.
- **Smaller storage** needs (gigabytes)
- **Millisecond response time**

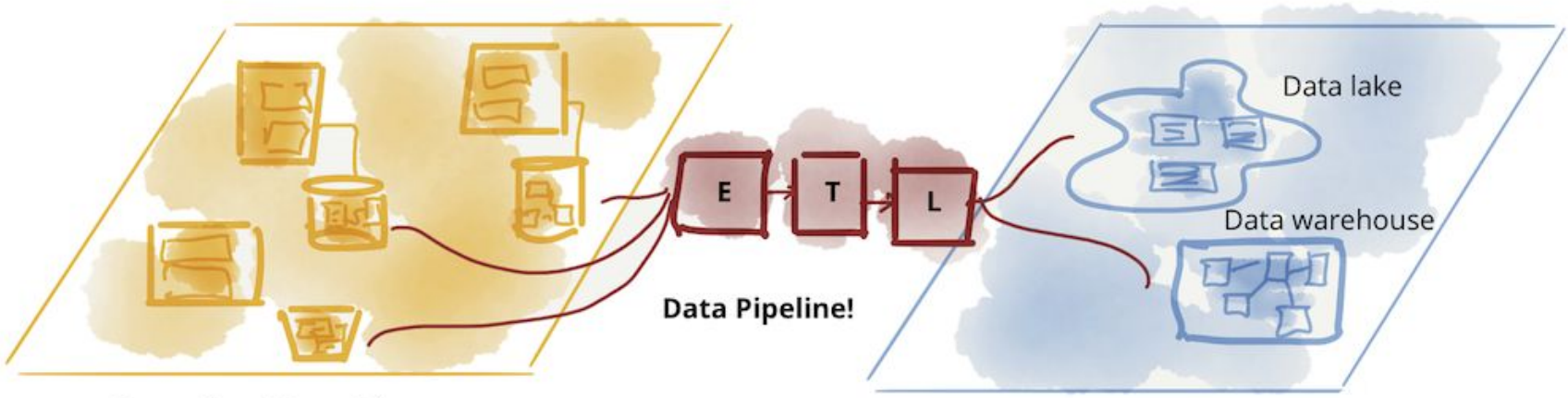
OLAP

- Analyzes large volumes of **historical and aggregated data** from **multiple sources** to support decision-making.
    - Ex: Trend analysis, customer behavior prediction, profitability insights.
- Frequently use **multidimensional** (cubes) or relational databases.
- Uses **star schema**, **snowflake schema**, or other analytical models.
- **Large storage needs** (terabytes to petabytes).
- **Response time in seconds or minutes**

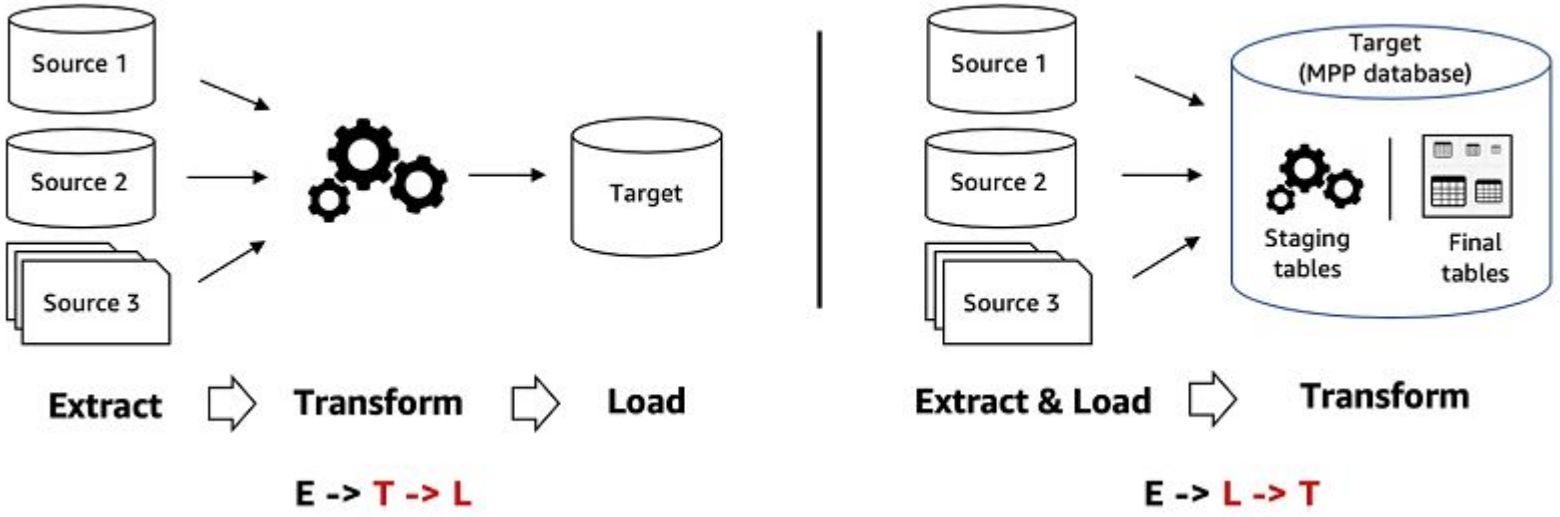
Definitions

- **Traditional**
  - **Data Warehouse:** Centralized storage for structured, processed data optimized for analytics.
  - **Data Mart: Subset of a data warehouse focused on specific business areas.**
- **Modern**
  - **Data Lake:** Repository for raw, unstructured, and structured data at scale.
  - **Data Lakehouse:** combines the benefits of data lakes and data warehouses into a single platform

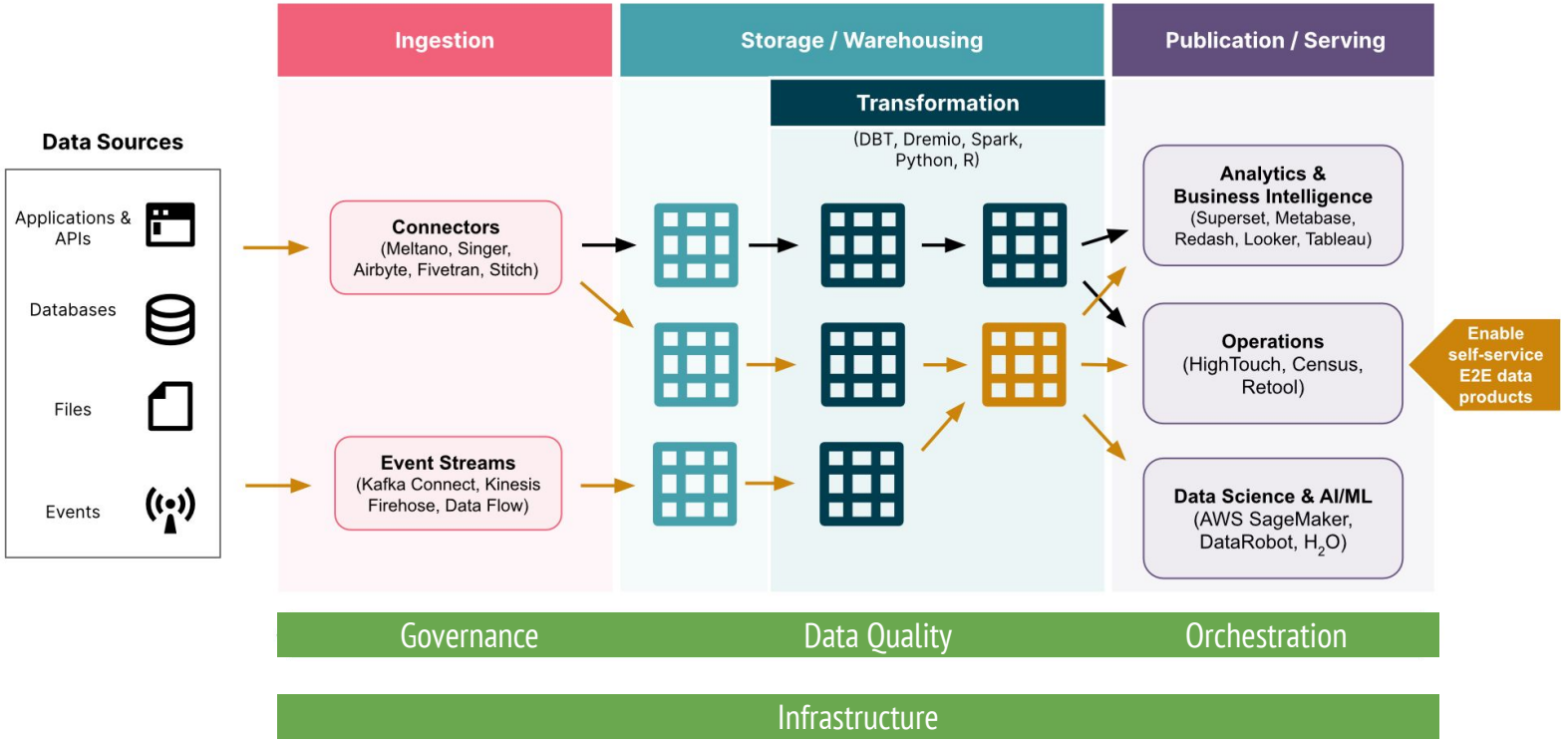
How they fit together



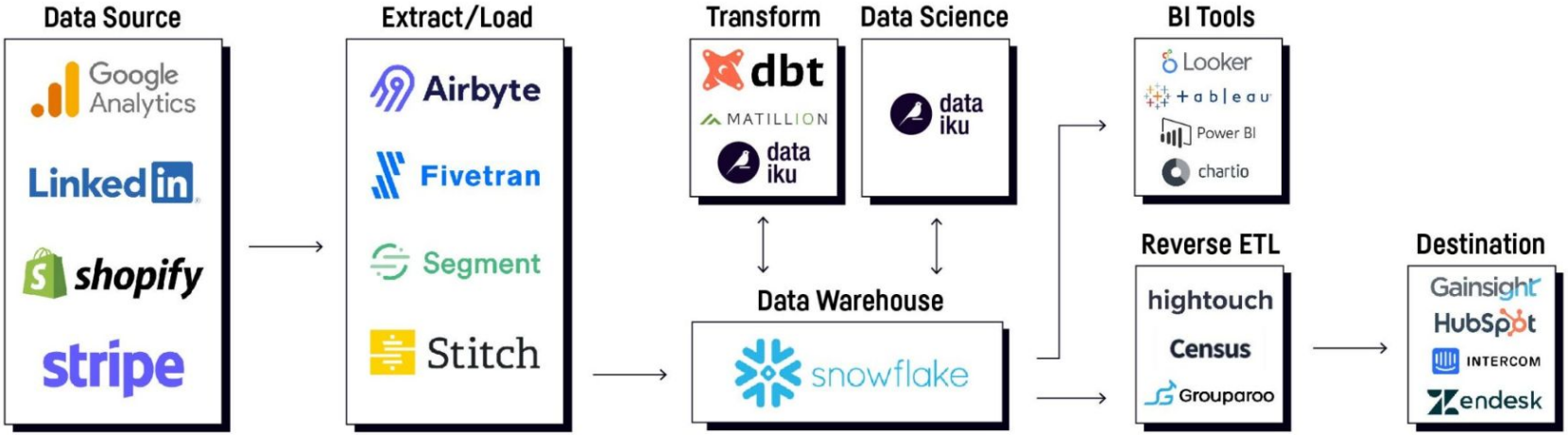
ETL vs ELT


Source 1
Source 2
Source 3
Target
Extract
Transform
Load
E -> T -> L
Source 1
Source 2
Source 3
Target
(MPP database)
Staging
tables
Final
tables
Extract & Load
Transform
E -> L -> T

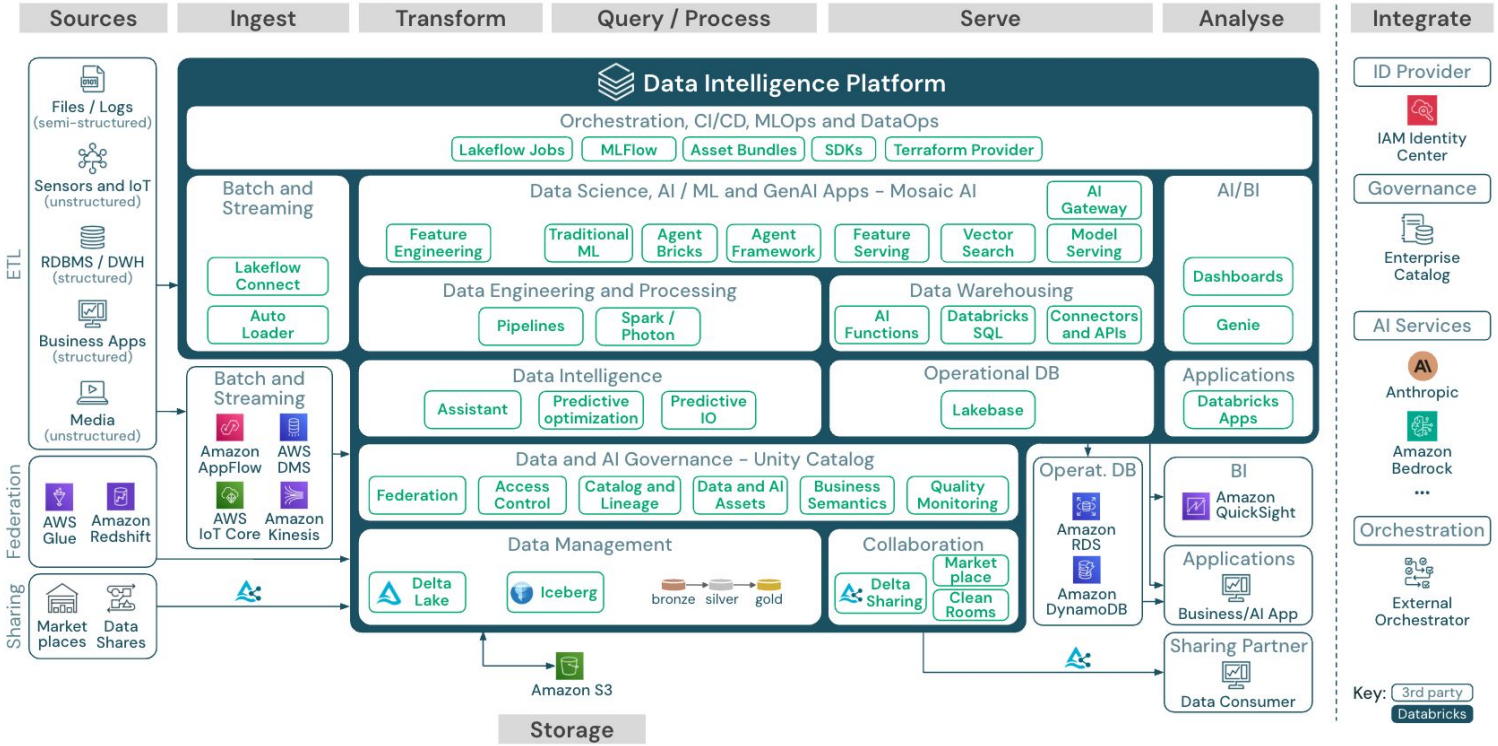
What is a data architecture?



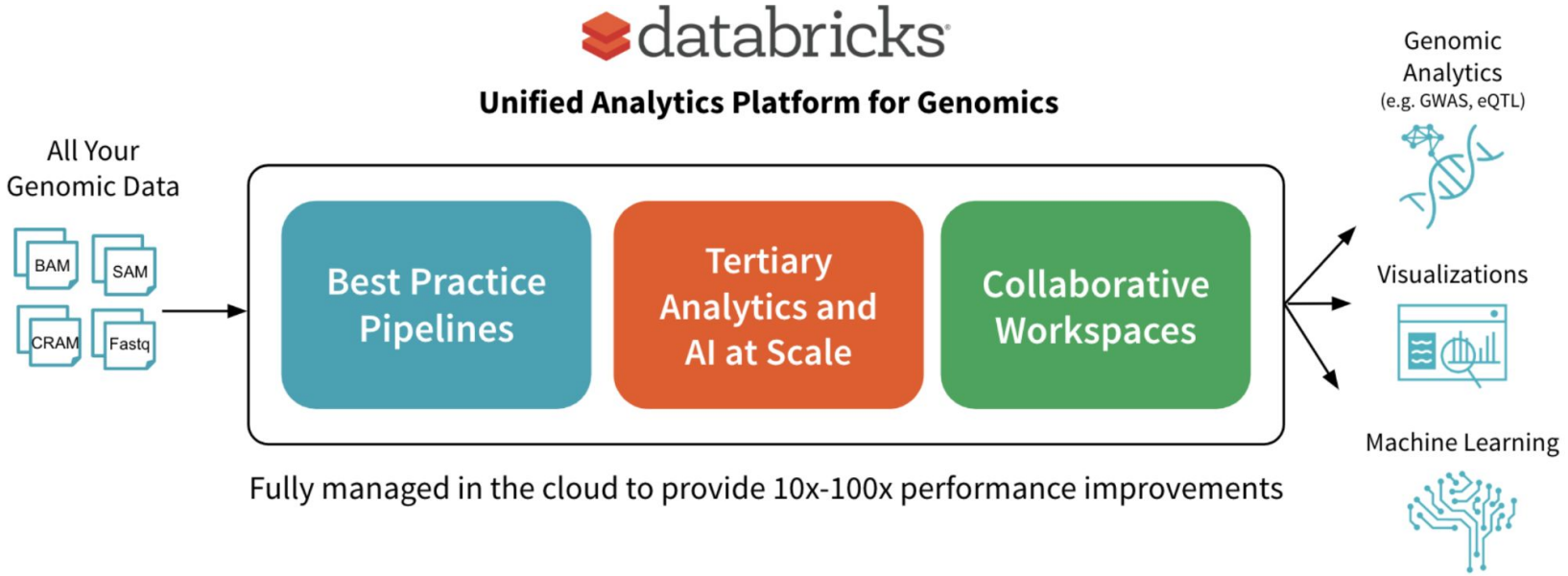
The Modern Data Platform



The Modern Data Platform: Databricks

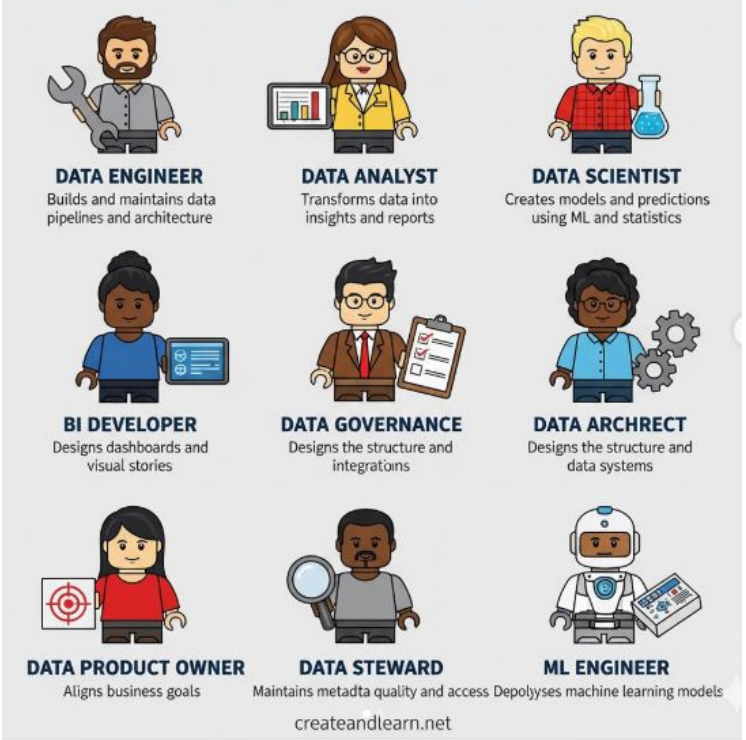


The Modern Data Platform: Genomics



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4. **Orchestration**

Introduction (1)

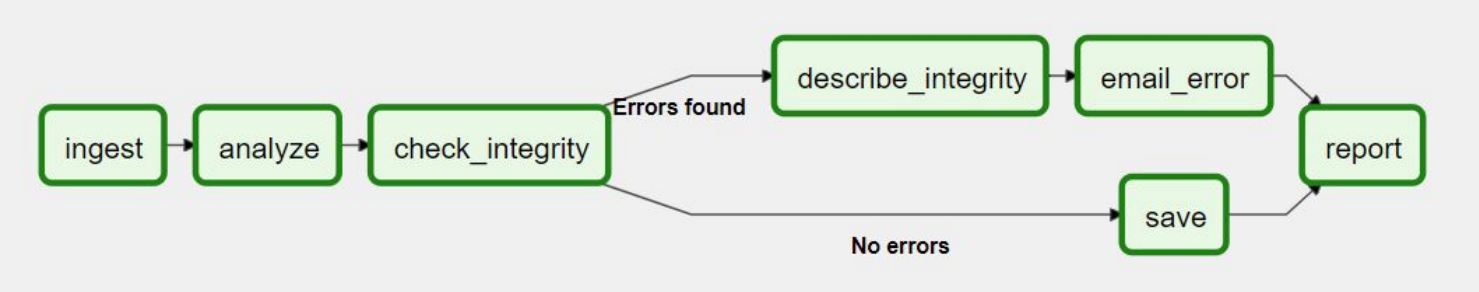
- Definition: **orchestrated** sequence of steps to move/transform data.
- Motivation (Similar to CI/CD)
  - Eliminates manual data handling.
  - Ensures data consistency and quality.
  - Speeds up analytics and AI pipelines.
  - Enables scalability across diverse data sources.
- Conceptual example: app event → analytic table → dashboard.

Introduction (2)

- Components
  - Data Ingestion: collecting data from multiple sources.
  - Transformation: cleaning, structuring, enriching.
  - Scheduling & Workflow Management: defining dependencies and timing.
  - Monitoring: tracking job status and data health.
- Common Tools & Technologies
  - **Apache Airflow**, Prefect, Dagster – workflow orchestration.
  - AWS Step Functions, AWS Glue Workflows, Azure Data Studio, Databricks Jobs – infrastructure-level orchestration.
  - dbt – transformation orchestration.
  - Integration with data lakes, warehouses, and BI tools.

Main orchestration abstraction: The DAG

- A DAG (Directed Acyclic graph) is a model that encapsulates everything needed to execute a workflow:
  - Schedule: When the workflow should run.
  - Tasks: tasks are discrete units of work that are run on workers.
  - Task Dependencies: The order and conditions under which tasks execute.
  - Callbacks: Actions to take when the entire workflow completes.
  - Additional Parameters: And many other operational details.
- Desirable properties
  - idempotency
  - retries (with backoffs)
  - observability

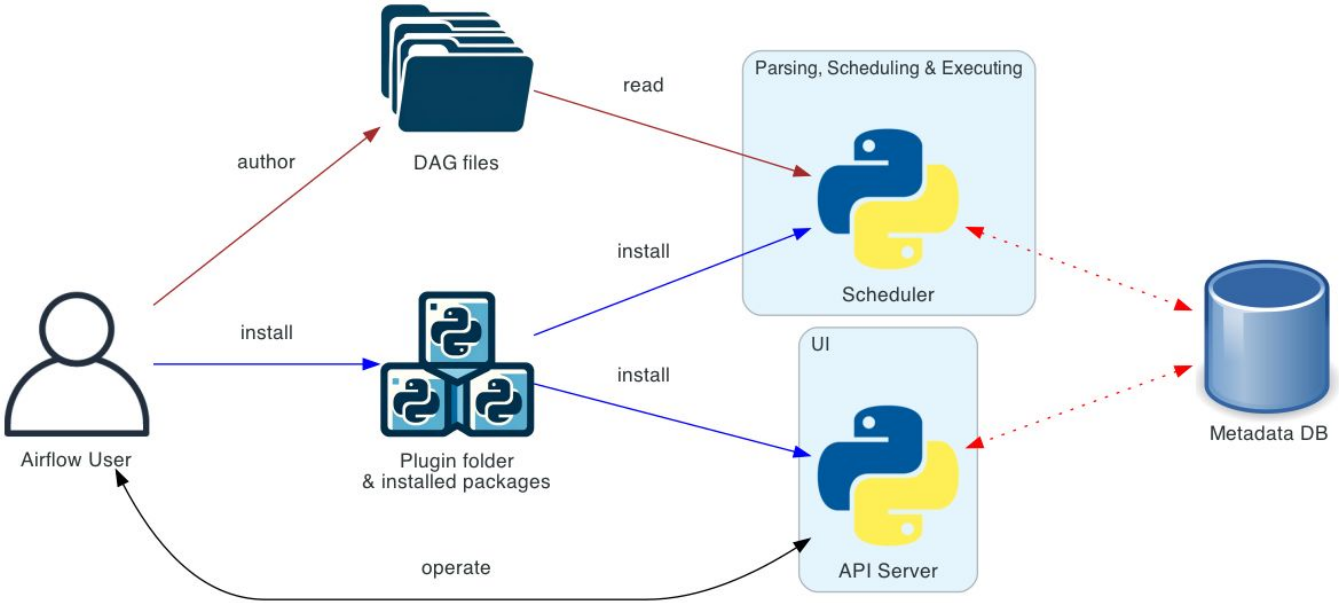


Standard tool: Apache Airflow



- Is a Python based workflow orchestration platform
  - Workflows are defined with standard Python code
  - Modular and easy to extend with many integrations with other services.
  - Open source and can be run locally
  - Has managed versions in most cloud providers

Airflow: Architecture



Airflow UI

https://airflow.apache.org/docs/apache-airflow/stable/ui.html

Airflow UI: How are DAGs defined

- https://airflow.apache.org/docs/apache-airflow/stable/index.html
- https://airflow.apache.org/docs/apache-airflow/stable/tutorial/fundamentals.html#understanding-operators

Dudas?